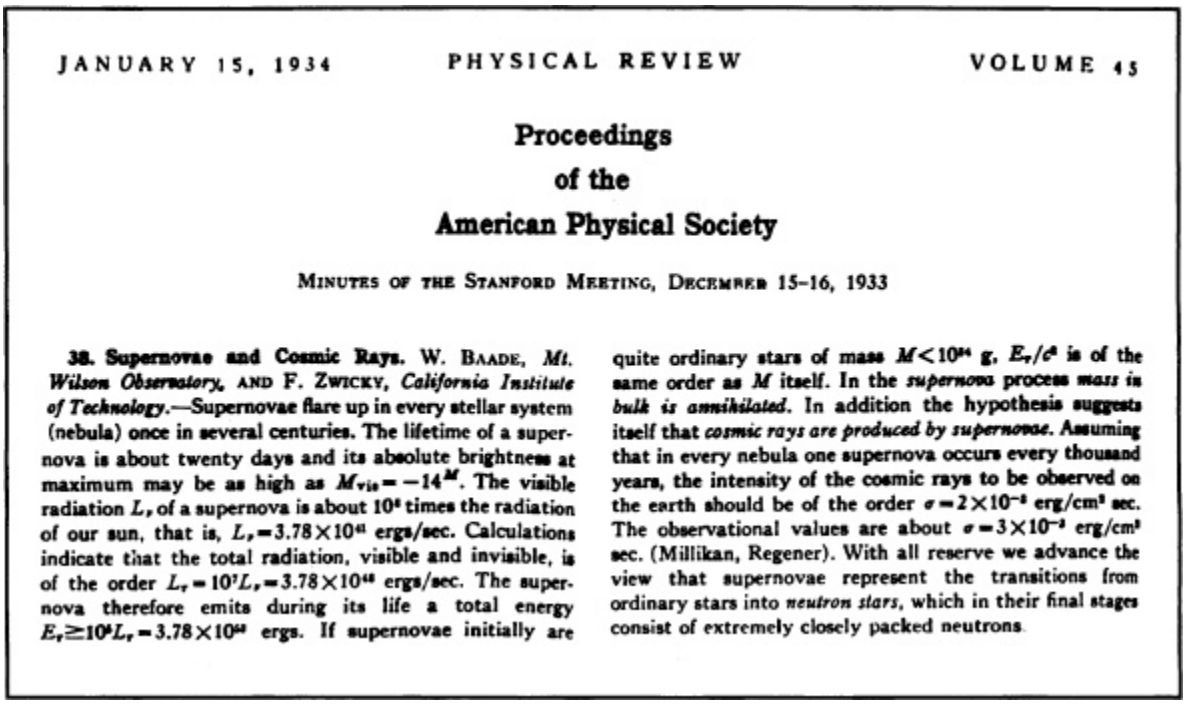


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MINUTES OF THE STANFORD MEETING, DECEMBER 15–16, 1933

38. Supernovae and Cosmic Rays. W. BAADE, *Mt. Wilson Observatory*, AND F. ZWICKY, *California Institute of Technology*.—Supernovae flare up in every stellar system (nebula) once in several centuries. The lifetime of a supernova is about twenty days and its absolute brightness at maximum may be as high as $M_{vis} = -14^{M}$. The visible radiation L_r of a supernova is about 10^8 times the radiation of our sun, that is, $L_r = 3.78 \times 10^{41}$ ergs/sec. Calculations indicate that the total radiation, visible and invisible, is of the order $L_r = 10^7 L_r = 3.78 \times 10^{48}$ ergs/sec. The supernova therefore emits during its life a total energy $E_r \geq 10^8 L_r = 3.78 \times 10^{49}$ ergs. If supernovae initially are quite ordinary stars of mass $M < 10^{34}$ g, E_r/c^2 is of the same order as M itself. In the *supernova* process *mass in bulk is annihilated*. In addition the hypothesis suggests itself that *cosmic rays are produced by supernovae*. Assuming that in every nebula one supernova occurs every thousand years, the intensity of the cosmic rays to be observed on the earth should be of the order $\sigma = 2 \times 10^{-3}$ erg/cm^2 sec. The observational values are about $\sigma = 3 \times 10^{-3}$ erg/cm^2 sec. (Millikan, Regener). With all reserve we advance the view that supernovae represent the transitions from ordinary stars into *neutron stars*, which in their final stages consist of extremely closely packed neutrons.

5.2 Abstract of the talk on supernovae, neutron stars, and cosmic rays given by Walter Baade and Fritz Zwicky at Stanford University in December 1933.

Astronomers in the 1930s responded enthusiastically to the Baade—Zwicky concept of a supernova, but treated Zwicky's neutron-star and cosmic-ray ideas with some disdain. "Too speculative" was the general consensus. "Based on unreliable calculations," one might add, quite correctly. Nothing in Zwicky's writings or talks provided more than meager hints of substantiation for the ideas. In fact, it is clear to me from a detailed study of Zwicky's writings of the era that he did not understand the laws of physics well enough to be able to substantiate his ideas. I shall return to this later in the chapter.

Some concepts in science are so obvious in retrospect that it is amazing nobody noticed them sooner. Such was the case with the connection between neutron stars and black holes. Zwicky could have begun to make that connection in 1933, but he didn't; it would get made in a tentative way six years later and definitively a quarter century later. The tortured route that finally rubbed physicists' noses in the connection will occupy much of the rest of this chapter.